Exercise 6 :

a) $g(c) = E[(Y-c)^2]$

$= E[Y^2] - 2cE[Y] + c^2$ by E properties

$g'(c) = -2E[Y] + 2c$ setting it to zero yields

$g'(c) = 0 \iff \boxed{c = E[Y]}$ moreover $g''(c) = 2 > 0$ so it clearly a minimum.

b)

$E[(Y-h(X))^2] = E[E[(Y-h(X))^2 | X]$ let $\hat{Y} = E[Y|X]$

$E[(Y-\hat{Y}+\hat{Y}-h(X))^2|X] = E[(Y-\hat{Y})^2|X] + 2E[(\hat{Y}-h(X))^2|X] + 2E[(Y-\hat{Y})(\hat{Y}-h(X))|X]$

$E[Y-\hat{Y}|X] = E[Y|X] - \hat{Y} = \hat{Y} - \hat{Y} = 0$

$E[(Y-h(X))^2] = \underbrace{E[(Y-E[Y|X])^2]}_{\geq 0} + E[(E[Y|X]-h(X))^2]$

so the minimum is reached when $\boxed{h(X) = E[Y|X]}$